

A 0.5-V 125-MHz 256-Kb 22-nm SRAM With 10-aJ/bit Active Energy and 10-pW/bit Shutdown Power

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Abstract—Conventional low-voltage (LV) static random access memories (SRAMs) utilizing separate read-and-write assist circuits sacrifice access speed too much, leading to poor energy efficiency. Overlaying additional assist circuits to enable power-saving modes can exacerbate speed loss and energy inefficiency. This work proposes a coordinated read-write-hold-retention-shutdown (SD) assist circuit design for SRAMs to improve speed at low VDD and reduce leakage current in active and power-saving modes. This leads to overall power reduction and enhanced energy efficiency. The implemented 22-nm 256-Kb SRAM can operate down to 0.45 V, with the minimum energy point at 0.5 V. At 0.5 V, TT corner, and 25 °C, the measured maximum frequency ($f_{\max}$) is 125 MHz with a 10 aJ/bit active energy, representing a 6.25 higher frequency with a 91% energy reduction, compared to the state-of-the-art 0.5-V SRAM in 28-nm bulk CMOS. The SRAM has a 12 pW/bit deep-sleep power and a 10 pW/bit SD power at 0.5 V.

Index Terms—Assist circuits, deep sleep (DS), energy efficiency, low voltage (LV), power saving, shutdown (SD).

I. INTRODUCTION

READ-AND-WRITE assists are critical design techniques for lowering the supply voltage VDD of SRAMs without causing read, write, and half-select failures. The word-line (WL) voltage underdrive (WLUD) [1] is the primary concept of the read assist. The suppressed WL and the stepped WL are two common realizations based on the WLUD. However, the WLUD schemes reduce the bit cell's read current due to the weakened strength of pass gates, which results in degraded read speed.

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Three primary write-assist techniques are boosting the WL voltage above VDD, generating a negative bitline (BL) voltage, and lowering the VDD for bit cells. The boosted-WL (BWL) [2] scheme contradicts the WLUD scheme and degrades the read stability for row-wise half-select cells. The negative BL (NBL) [3] scheme needs high area and power overheads because of the required pumping capacitors and mechanism. The cell-VDD-lowering (CVDDL) [4] scheme loses power-reduction efficiency owing to the voltage-division-induced direct current (dc) in the CVDDL circuit.

Scaling VDD to the near-threshold regime presents significant challenges for low-voltage (LV) static random access memory (SRAM) design, including increased speed loss, degraded energy efficiency, and amplified variation effects. To improve read and write noise margins, designers often resort to complex bit cell structures with more than six transistors (6T), which increase cell area and risks using bit cells that are unavailable from foundries [5], [6], [7], [8]. While the SRAM in [9] utilized a foundry-provided 6T bit cell with 4-bit short bitlines (BLs) to operate at 0.5 V, the significant area overhead from utilizing 4×2 sub-arrays is a major drawback. The maximum frequency ($f_{\max}$) of this SRAM drops to only 20 MHz at 0.5 V. The SRAM in [10] adopted a new foundry-cell compatible dual-split-control 6T (DSC-6T) bit cell to reduce the area cost and achieve a high-speed LV operation. Although this SRAM successfully operates at a $f_{\max}$ of 400 MHz at 0.6 V utilizing an ideal and stable off-chip assist voltage, the proposed on-chip charge-sharing-based cell-VSS (CVSS) generation (CSVG) scheme was not verified. Our evaluation indicates that if the SRAM runs with the CSVG, the $f_{\max}$ at 0.6 V falls below 200 MHz at the typical PVT condition, and the value of the minimum VDD for total functionality rises.

In addition to the need for LV designs for reducing active power consumption, the SRAM designs for IoT endpoint applications especially demand high energy efficiency in active operations and, at best, have the support of various power-saving modes such as standby (SB), deep sleep (DS) [11], and shutdown (SD). For nominal-VDD SRAMs, diode-based CVDDL or CVSS raising (CVSSR) circuits [12] are vital circuits to reduce the effective VDD of SRAM cells in data-retained SB or sleep modes, and the power gating [13] for the whole SRAM macro can cut off the leakage current in the data-lost SD mode. The series resistance of diodes or

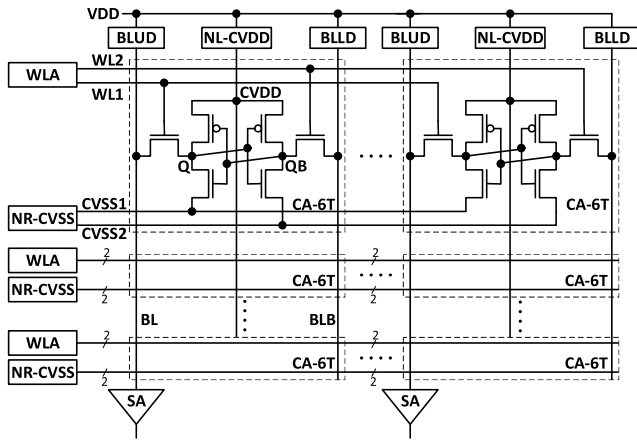


Fig. 1. Circuit of one bank of proposed LV SRAM.

power switches reduces the voltage headroom, leading to increased speed loss in LV SRAMs. The $f_{\max}$ of the 256-Kb SRAM in [9], equipped only with a SB mode, is down to 20 MHz at 0.5 V, as described before. Accordingly, we can infer that the $f_{\max}$ of the 0.6-V 256-Kb SRAM in [10] will be reduced further to much below 200 MHz if adding on power-saving designs with the on-chip CSVG.

This work proposes coordinated read-write-hold-retention-SD assist circuits for the SRAM to elevate speed at a low VDD, suppress leakage in active and power-saving, and achieve overall power reduction and energy efficiency enhancement. At 0.5 V/TT/25 °C, the implemented 22-nm 256-Kb SRAM has a 125-MHz $f_{\max}$ with 10 aJ/total-bit (the unit is also abbreviated as aJ/bit in this article) active energy, 12 pW/bit deep-sleep power, and 10 pW/bit SD power.

We organize the remainder of this article as follows. Section II introduces design concepts and operation principles. Section III analyzes noise-margin requirements. Section IV describes assist circuits. Section V presents the results of the test chip's implementation. Finally, Section VI concludes this article.

II. BASIC DESIGN CONCEPTS AND OPERATION PRINCIPLES

A. Basic Design Concepts

Fig. 1 illustrates the conceptual circuit of one bank of the proposed LV SRAM, adopting the proposed coordinated read-write-hold-retention-SD assist circuits. The following are the basic design concepts.

- 1) The adopted bit cell has separated BLs, WLs, and VSSs. All terminals, including the CVDD, are connected to the coordinated assist circuits. Hereafter, we call the bit cell a coordinated assist 6T (CA-6T) bit cell.
- 2) A normally lowered CVDD (NL-CVDD) assist circuit connects to the CVDD and a normally raised CVSS (NR-CVSS) assist circuit connects to split CVSSs. A timing-controlled WL assist (WLA) circuit connects to split WLs, and two voltage-controlled BL assist circuits connect to BL and complementary BL (BLB).

These assist circuits coordinate across active and power-saving modes.

- 3) The CA-6T cell has the same layout style as the DSC-6T cell [10] and has the same cell area as the DSC-6T and the foundry-provided standard 6T cells.
- 4) Many SRAMs, including the DSC-6T-based SRAM, utilized the NBL as the write assist, with a high area and power overhead. This work employs a no-dc and fast-switching NL-CVDD circuit that is improved from that presented in [14].
- 5) The hold noise margin typically exceeds the read noise margin (RNM), suggesting that reducing the CVDD of inactive banks can significantly reduce leakage current. Conventional SRAMs do not exploit this opportunity. Our proposed column-wise no-dc NL-CVDD circuit suppresses leakage current in un-accessed cells while assisting written cells. The NL-CVDD circuit's fast-switching feature prevents read speed degradation.
- 6) The NR-CVSS circuit is a diode-based fast-switching circuit. Unlike the VSS voltage in [10], which is at the ground voltage when not in read/write assist modes, this work adopts the NR-CVSS circuit to raise the VSS voltage of un-accessed cells. Therefore, the NR-CVSS circuit further reduces leakage and provides read/write assistance for the CA-6T-based SRAM.
- 7) Unlike a global CSVG circuit in the DSC-6T-based SRAM [10], the NR-CVSS circuit is row-based to improve operation speed and increase variation resiliency.
- 8) The WLA circuit solely controls the timing of split WLs without requiring any voltage boosting or suppression.
- 9) SRAMs usually utilize PMOS-based BL loads (BLLDs) for BL and BLB. Here, we adopt the conventional BLLD for BLB and propose a BL under-drive (BLUD) circuit for BL to assist the single-ended dynamic sense amplifier (SA) in improving access speed and the RNM, as described later.
- 10) To be described later, the proposed NL-CVDD circuit has a stacked-PMOS circuit structure and can adapt into a super-cut-off switch [13] for leakage suppression in the SD mode.

We will describe the operation and coordination principles of the proposed assist circuits in the following.

B. Operation Principles

There are five operation modes: read, write, SB, DS (or called retention), and SD (or called power down). Fig. 2 illustrates the operation waveforms, and we describe the operation principles as follows.

- 1) *Write Operation*: Cycle 1 performs a write operation. CVDDs of written and un-accessed cell banks are lower than VDD by ΔVC , and the supply voltage PVDD of peripheral circuits is equal to VDD. CVSS1 and CVSS2 of un-accessed banks are higher than that of ground voltage (GND) by ΔVS . CVSS1 and CVSS2 of the accessed row are pulled down from $GND + \Delta VS$ to GND in the positive and negative phases, respectively, and WL1 and WL2 are asserted in the

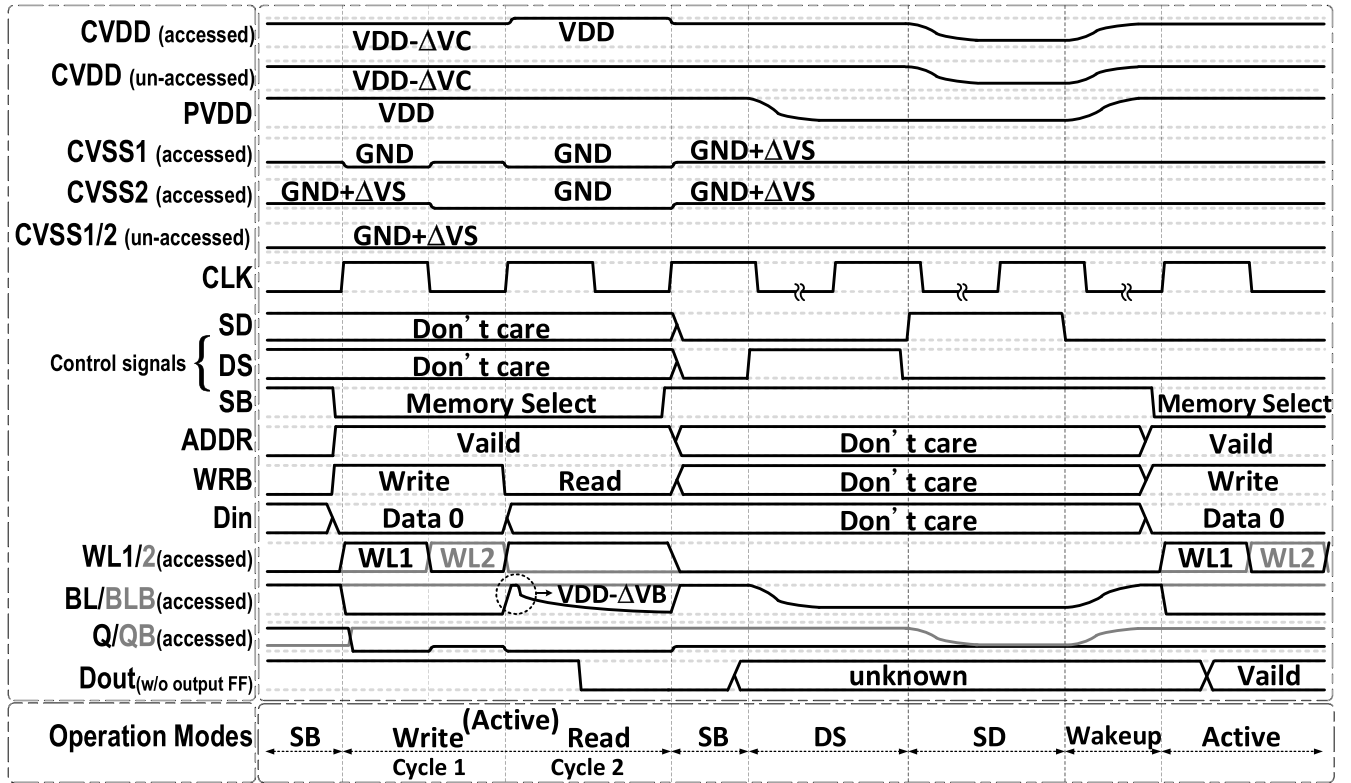


Fig. 2. Operation waveforms of proposed LV SRAM.

positive and negative phases, respectively. CVSS1 and CVSS2 of un-accessed rows keep at ΔVS above GND, whether in the positive or negative phases. Take writing a “0” as an example. We should pull the BL voltage to GND and keep the BLB voltage at VDD. Hence, during the positive phase, the voltage at Q goes to GND, and that at QB rises from GND + ΔVS to VDD- ΔVC to complete the operation. In the negative phase, CVSS1 returns to GND + ΔVS to push the voltage at Q up to GND + ΔVS . CVSS2 goes to GND and WL2 to VDD in the negative phase. Since the write is complete in the first phase, the logic values of Q and QB are unchanged in the second phase. Note that all un-accessed cells are in a low-leakage state, with the CVDD kept at VDD- ΔVC and CVSS1 and CVSS2 at GND + ΔVS .

2) *Read Operation*: Cycle 2 performs a read through the BL, with WL1 asserted high and WL2 kept low during the whole cycle. The CVDD of the accessed bank goes to VDD, and the others at VDD- ΔVC . The CVSS1 of the accessed row goes to GND, but CVSS2 keeps at GND + ΔVS . CVSS1 and CVSS2 stay at GND + ΔVS for all un-accessed rows. The PVDD is still VDD. Initially, the voltage at BL is lowered from VDD by ΔVB , controlled by the BLUD circuit, to increase the RNM and read speed. After asserting WL1, the voltage at BL is affected by that of Q. If reading a “0,” the BL voltage drops to GND. All un-accessed cells are in a low-leakage state, whether in accessed or un-accessed banks. CVSS1 and CVSS2 of un-accessed rows in all banks are GND + ΔVS , and CVDD of all un-accessed banks is VDD- ΔVC for reducing the total read energy.

3) *Standby*: For the example presented in Fig. 2, the SRAM enters the SB mode after pulling the control signal SB high. Although the system clock signal keeps running in this mode,

the SRAM turns off all input flip-flops. Moreover, the PVDD is at VDD, but the super-cut-off switches power off WL drivers. Note that CVDD and CVSS of all banks are at VDD- ΔVC and GND + ΔVS , respectively, to achieve higher power saving than conventional SRAMs.

4) *Deep Sleep*: The SRAM enters DS by pulling up signal DS. This mode reduces the power consumption below that in SB if the system does not need to access the SRAM for a long time. The system design demands that the SRAM returns quickly to active mode. By the NL-CVDD and the NR-CVSS, the cell banks are already low-leakage, low-power designs without extra quick wakeup designs. To achieve the goal of lower power, the SRAM powers down all the peripheral circuits, including the BLLDs, by utilizing the super-cut-off switches. The proposed SRAM can wake up from the DS within one clock cycle.

5) *Shutdown*: The SRAM enters the SD mode by pulling up the SD signal. The SRAM does not need to retain the cell contents in this mode. Consequently, the SRAM supercuts off all cell arrays in addition to peripheral circuits. The proposed SRAM can wake up from the SD mode within three clock cycles.

III. ANALYSIS OF NOISE-MARGIN REQUIREMENTS

The foundry provides 22-nm 0.6 V $\pm$ 10% and -40° – 125° °C device models, and the 0.5-V design is also on this model with accuracy loss. We adopted the following steps to simulate noise margins for design concept proving. First, we performed static RNM and write margin (WM) by 1-million (1M) points Monte Carlo (MC) simulations to check the bit cell’s robustness over 0.6 V $\pm$ 10%, -40° – 125° °C, and

process corners. Second, we performed 0.1M MC simulations at 0.5 V for the static RNM at FnSp and 3σ global corners, with 3σ local variations for both corners. We also similarly verified the WMs.

Fig. 3(a) and (b) presents the histograms of 1M static RNM MC simulation results of 0.54 V, 125 °C, and the process condition at the worst corner (FnSp) and the 3σ global corner, respectively. The μ/σ values are 6.77 and 6.92, respectively, and the minimum RNM values are 34 and 30 mV, respectively.

Fig. 3(c) and (d) presents the histograms of 1M static WM MC simulation results of 0.54 V, -40 °C, and the process condition at the worst corner (SnFp) with 3σ local variations and at the 3σ global corner plus 3σ local variations, respectively. The μ/σ values are 7.21 and 7.64, respectively, and the minimum WM values are 78 and 107 mV, respectively.

For 0.5-V operation with accuracy loss of device models, we performed only 0.1M static RNM and WM MC simulations and only reported the results of the RNM. Fig. 4(a) displays the histogram at the FnSp corner with 3σ local variations, and the μ/σ and minimum RNM values are 6.09 and 29 mV, respectively. Fig. 4(b) presents the histogram at the 3σ global corner plus 3σ local variations, and the μ/σ and minimum RNM values are 6.29 and 25 mV, respectively.

After static noise margin analysis, we conduct the dynamic noise margin analysis to determine allowable ΔVS , ΔVB , and ΔVC ranges for reliable assistance. We performed 0.1M-point 0.5-V transient MC simulations with 0–250 mV noise added to the bit cell with a step of 10 mV, as described below.

A. Simulations for Minimum ΔVS and ΔVB

We conducted MC post-layout simulations at the condition of FnSp/0.5 V/125 °C for a 128×4 cell array with four-column interleaving to obtain the dynamic RNM (DRNM) of a cell in read, according to [3]. Simulation results presented in Fig. 5 indicate that even if $\Delta VB = 0$ V, the μ/σ value of DRNM is higher than six, and even the minimum ΔVS is only 10 mV. We set the minimum ΔVB to 0 V; even a larger ΔVB increases the μ/σ . Increasing the minimum ΔVS enhances DRNM, improves the dynamic WM, and reduces power consumption.

B. Simulations for Minimum ΔVC

We evaluate the minimum ΔVC by simulation of the dynamic WM (DWM), according to [15], for a cell in the write mode. We set ΔVS to be 10 mV according to the analysis of DRNM and conducted MC post-layout simulations at SnFp/0.5 V/-40 °C. Fig. 6 presents the results, indicating that a ΔVC of at least 30 mV achieves a 100% successful write rate and achieves the μ/σ value of DWM ≥ 6 . Although increasing ΔVS lowers ΔVC for the same DWM, this work sets the minimum ΔVC as 30 mV.

C. Simulations for Maximum ΔVS

The maximum ΔVS affects the hold noise margin. We simulate a circuit in the write mode to obtain the column half-select cell's dynamic hold noise margin (DHNM)

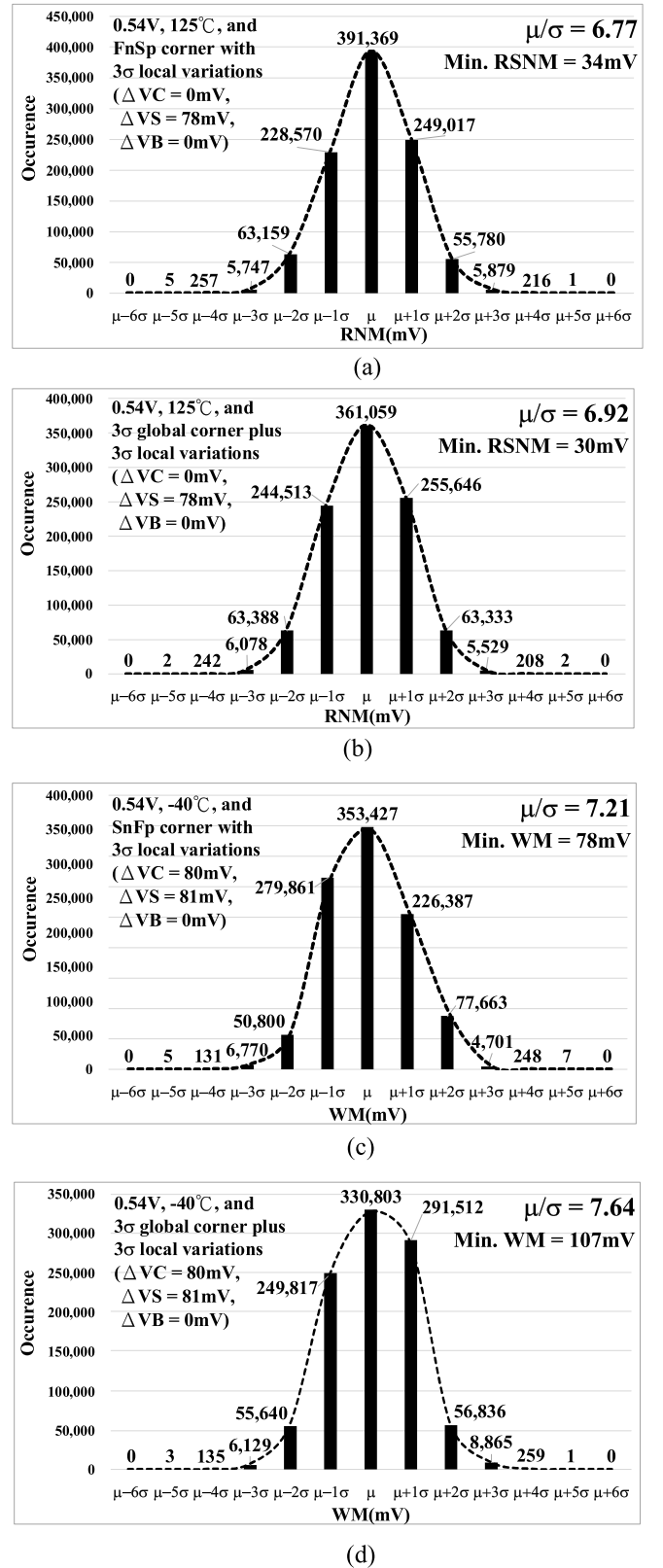


Fig. 3. 1M MC simulations of static RNM at 0.54 V, 125 °C and (a) FnSp with 3σ local variations and (b) 3σ global corner plus 3σ local variations. 1M MC simulations of static WM at 0.54 V, -40 °C, and (c) SnFp with 3σ local variations and (d) 3σ global corner plus 3σ local variations.

by setting ΔVC to 30 mV. We conduct MC post-layout simulations at the worst condition of FnSp/0.5 V/125 °C.

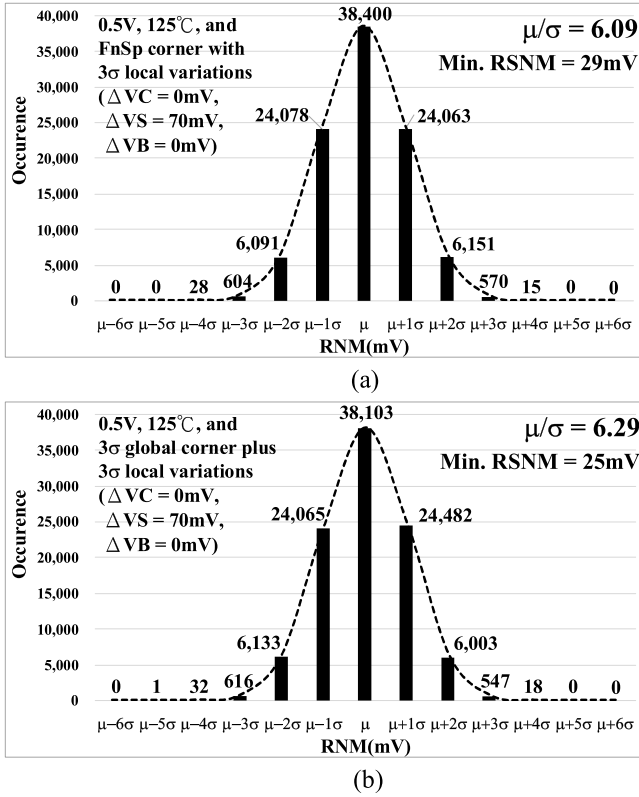


Fig. 4. 0.1M MC simulations of static RNM at 0.5 V, 125 °C, and (a) FnSp corner with 3σ local variations and (b) 3σ global corner plus 3σ local variations.

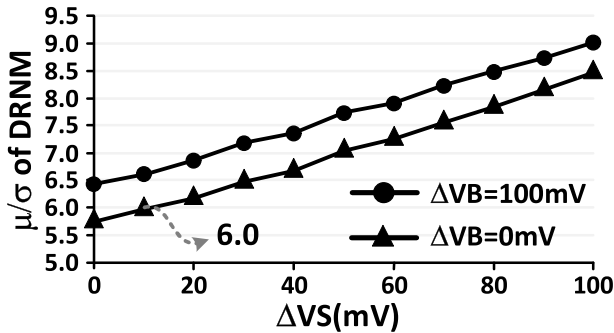


Fig. 5. Simulation results of DRNM.

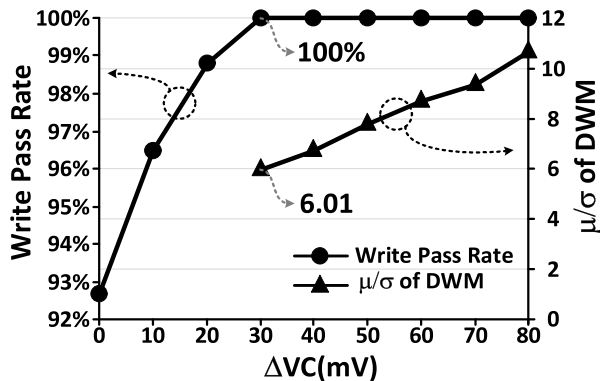


Fig. 6. Simulation results of DWM.

Fig. 7 presents simulation results. The maximum ΔVS is 170 mV for achieving the μ/σ value of DHNM ≥ 6 .

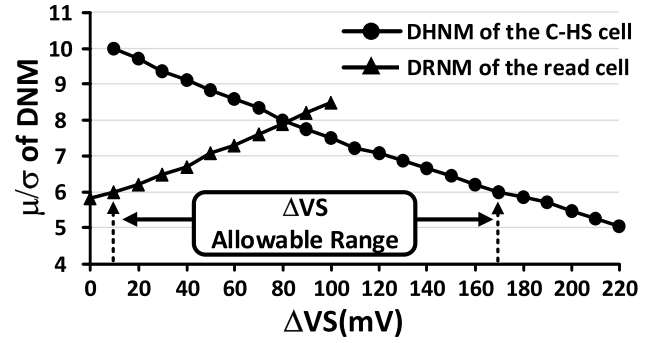


Fig. 7. Simulation results of DHNM and DRNM.

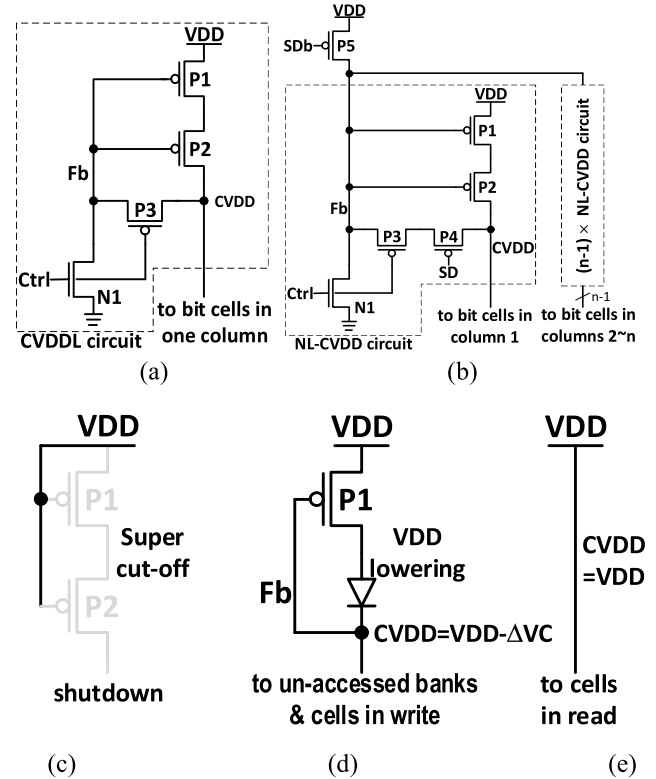


Fig. 8. (a) CVDDL and (b) NL-CVDDL circuits. Equivalent circuits of NL-CVDDL circuit for (c) SD, (d) un-accessed, and (e) read modes.

IV. ASSIST CIRCUITS

A. NL-CVDDL Circuit

Fig. 8(a) presents the original CVDDL circuit in [14] for the write assist only. Fig. 8(b) illustrates this work's proposed NL-CVDDL circuit for write, hold, retention, and SD assist. P1–P4 and N1 construct a sub-circuit for serving one column of cells, and all columns in an SRAM bank share one P5. Complementary signals SD and SDb control P4 and P5.

In SD mode, SDb and Ctrl go low to turn on P5, pulling the voltage at node Fb high to turn off P1 and P2 to achieve a super cut-off effect for significant leakage suppression. For control purposes only, P5 can be a minimum-sized device. The equivalent circuit for this mode is presented in Fig. 8(c).

When not in SD mode, SDb remains high, and the NL-CVDDL circuit functions similar to the CVDDL circuit. Signal Ctrl is low for a column of cells in write and other un-accessed banks. Fig. 8(d) indicates the equivalent circuit for

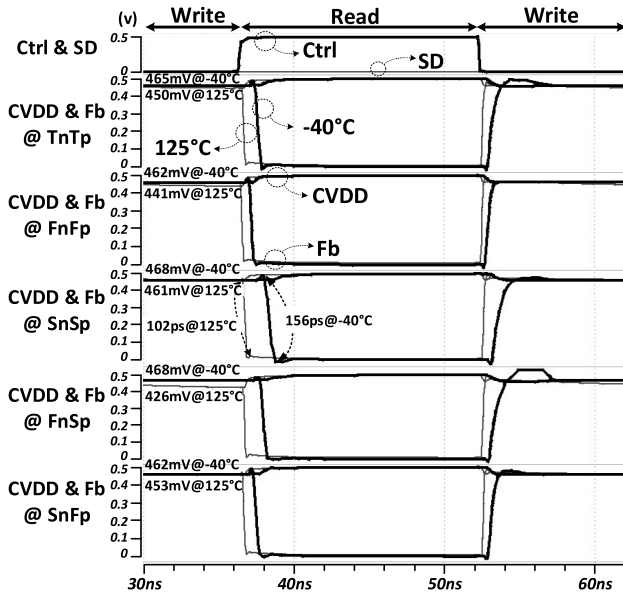


Fig. 9. Transient response of NL-CVDD circuit.

these scenarios. P3 and P4 make a short effect between nodes Fb and CVDD, and P2 acts as a diode to induce a voltage drop ΔVC . P3 and P4 feed CVDD back to control P1. The voltage of CVDD depends on the threshold voltage of P2 and the feedback effect. P1 is finally in the OFF-state, and leakage current flows through P1 and P2. The ΔVC suppresses the leakage current of un-accessed banks. Ctrl goes high only to pull the voltage at node Fb low during read mode. The equivalent circuit for this mode is illustrated in Fig. 8(e), where VDD supplies current to the accessed cells.

The voltage levels and switching speed of CVDD are crucial for the NL-CVDD circuit. Fig. 9 presents post-layout simulation waveforms of a 128-cell column under various operating conditions at a 0.5-V VDD. CVDD ranges from 426 to 468 mV across different conditions, meeting the ΔVC specification determined by DWM analysis. With the condition of SnSp and -40°C , the slowest falling time of the voltage of node Fb is only 156 ps.

B. NR-CVSS Circuit

Fig. 10 presents the proposed row-wise NR-CVSS circuits, stacking on a column-wise distributed diode array for n rows of cells in a bank. Every NR-CVSS circuit comprises dual sub-circuits to control CVSS1 and CVSS2 voltages. Each sub-circuit comprises a pass NMOS transistor to control the connection between the diode array and the CVSS line, a local diode circuit to create a voltage drop between CVSS and GND, a charging circuit to rapidly increase the CVSS voltage, and a discharging circuit to quickly decrease the CVSS voltage.

When not in SD mode and the cell is on SB, the globally distributed column-wise NMOS diodes and the local diode circuit cooperate to provide a CVSS voltage “VD” to increase variation resilience when the pass NMOS conducts. When required to pull down CVSS to GND, the SRAM turns N6 off and N1 on. When required to recharge the CVSS to VD, the SRAM turns N1 off, reconnects CVSS to the diode array,

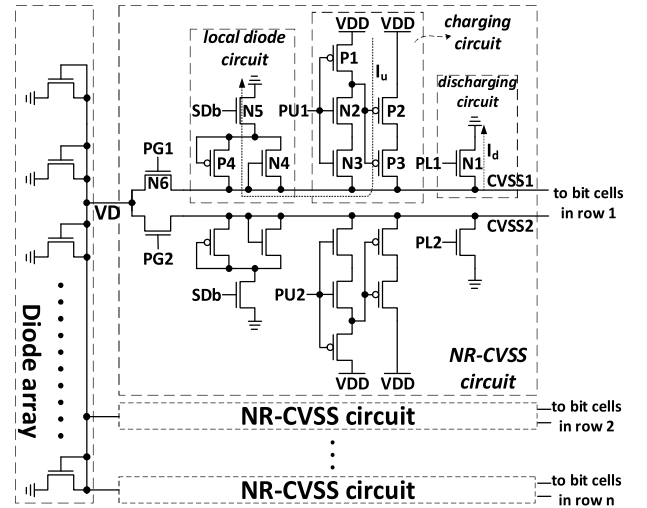


Fig. 10. Proposed NR-CVSS circuits and diode array.

TABLE I
CVDD AND CVSS AT DIFFERENT OPERATION CONDITIONS

Process	Temp.	CVDD (mV)	CVSS (mV)	CVDD-CVSS (mV)
TnTp	125 °C	450	86	364
	-40 °C	465	59	406
FnFp	125 °C	441	92	349
	-40 °C	462	65	397
SnSp	125 °C	461	72	389
	-40 °C	468	51	417
FnSp	125 °C	426	72	354
	-40 °C	468	53	415
SnFp	125 °C	453	101	352
	-40 °C	462	75	387

and enables the charging circuit. The structure of the fast charging circuit is similar to that of the NL-CVDD circuit. For accelerating but not overcharging, the pulsed signal PU1 goes high and turns P1 off and N2 and N3 on to make P2 and P3 act as a charging diode. Through N2 and N3, the raised CVSS has an effect of negative feedback to control P2 and P3 such that the charging behavior is variation resilient, similar to the NL-CVDD circuit. The local diode circuit also helps prevent overcharging. When entering SD mode, N6 and N5 are turned off to block all leakage paths to ground.

Table I presents the simulated values of CVDD and CVSS at 0.5 V with various process and temperature corners. CVDD is always below 470 mV, satisfying the DWM's requirement, and CVSS is between 51 and 101 mV, meeting the DRNM and DHNM's requirements. The smallest voltage headroom between CVDD and CVSS is 349 mV, about 70% of 0.5 V, sufficient for achieving power savings.

The CSVG scheme [10] needed measurement results to verify its feasibility. Fig. 11(a) presents the simulation results of our evaluation at 0.6 V/SnSp/125 °C. The globally controlled CSVG takes 1.13 ns to raise the CVSS voltage, becoming an obstacle to high-speed operation; thus, the $f_{\max}$ at 0.6 V falls much below 200 MHz. The NR-CVSS circuit has two features that solve this issue. First, it is row-based

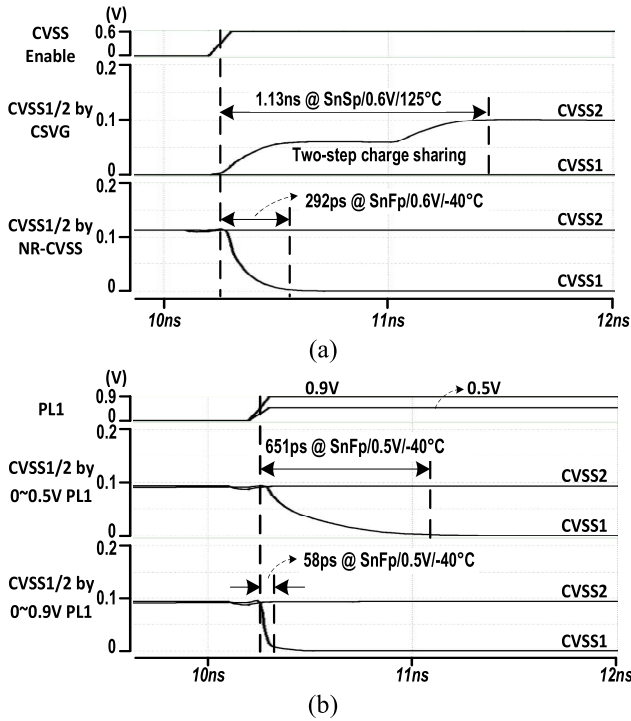


Fig. 11. (a) Comparison of CVSS generations and (b) effect of boosting control voltage.

and avoids a long transmission of signal ΔV_S across the whole cell array. Second, the NR-CVSS circuit utilizes the one-NMOS-only sub-circuit to pull down the ΔV_S . Fig. 11(a) also presents the simulation results of the NR-CVSS circuit at 0.6 V/SnFp/-40 °C, and the time for pulling down the CVSS voltage is reduced remarkably to 292 ps. When working at 0.5 V, the falling time of the CVSS voltage of the NR-CVSS circuit increases to 651 ps, as demonstrated in Fig. 11(b). We can apply a boosted voltage to increase the driving capability of N1, as the work [16] did. The falling time of the CVSS voltage reduces to 58 ps, as illustrated in Fig. 11(b), when the boosted voltage is 0.9 V, typically available in a 22-nm SoC.

C. BLUD and BLLD Circuits

The circuit in Fig. 12 includes BLLD and BLUD circuits, a column multiplexer, a single-ended SA, and signal DS/SD-controlled PMOS transistors for reading the stored data in a column of bit cells. We employ the DS/SD-controlled transistors to generate the super-cutoff effect in DS and SD modes. Two columns in neighboring banks share one SA. The BLUD has one more PMOS than the BLLD.

The BL and the data line (DL) are pre-charged to VDD initially and under-driven to $VDD - \Delta V_B$ at the beginning of the reading. Simulated waveforms at 125 MHz in Fig. 13 confirm that utilizing the BLUD earns a 36.4% speed improvement at 0.5 V/TT/25 °C.

V. TEST CHIP

We implemented a 22-nm 256-Kb SRAM test chip to verify the proposed design techniques. Fig. 14(a) displays the block diagram of the test circuit, and we can apply many kinds of

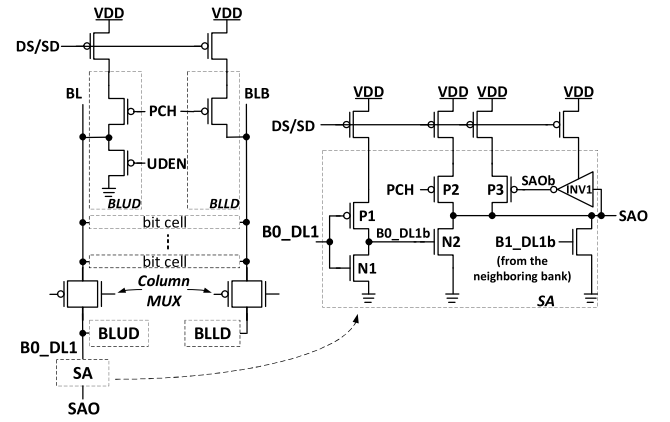


Fig. 12. Equivalent circuit of one column in a bank.

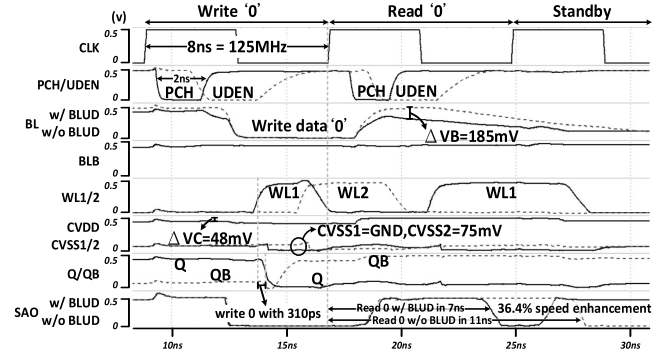


Fig. 13. Post-layout simulated waveforms of BL circuit at 125 MHz with an operation condition of 0.5 V/TT/25 °C.

TABLE II
EFFECTS OF LEAKAGE POWER REDUCTION

Assist voltages	CVDD	VDD	VDD	VDD- ΔV_C	VDD- ΔV_C
	CVSS	GND	GND+ ΔV_S	GND	GND+ ΔV_S
Leakage power (nA) (Normalized)	930.3 (1.00)	409.3 (0.44)	837.27 (0.90)	297.7 (0.32)	

test patterns, including checkerboards, to verify the function at various clock frequencies provided by the on-chip phase-locked loop. We also pull out data-out signals Dout [0] of the shortest path and Dout [31] of the most critical path for observing the waveforms at f_{max} . Fig. 14(b) presents the die photograph of the test chip, embedding a memory built-in self-test circuit (MBIST) for at-speed chip verification. The SRAM has eight banks, each containing two arrays of 128×128 cells. The size of the SRAM macro is $376.14 \times 172.16 \mu m$.

We evaluated the effect of leakage power reduction in several design scenarios. Table II presents post-layout simulation results of an array of 128×128 cells at 0.5 V/TT/25 °C. The proposed design utilizing NL-CVDD and NR-CVSS circuits can achieve a 68% leakage power reduction, leading to higher power saving in active, SB, and deep-sleep modes for an SRAM with a high memory capacity organized in many banks.

Table III presents the active power consumption of different blocks in the 256-Kb SRAM performing a read operation at 79.5 MHz without applying the BLUD technique. We can see a meager percentage of leakage from cell arrays due to the significant leakage reduction from utilizing the NL-CVDD and

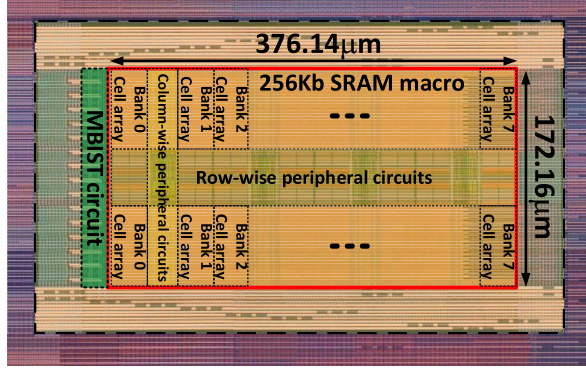
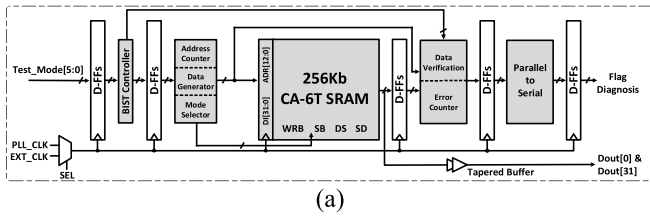


Fig. 14. (a) Block diagram and (b) die photograph of test chip.

TABLE III
EFFECTS OF ACTIVE POWER REDUCTION

Blocks	Cell arrays	Bit-line loads	Assist Circuits	Peripherals	Whole SRAM
Power(μ W) @ 79.5 MHz w/o BLUD (percentage)	17.37 (6.20%)	88.62 (31.61%)	0.71 (0.25%)	173.69 (61.95%)	280.39 (100%)
Power(μ W) @ 125 MHz w/ BLUD (percentage)	16.64 (5.88%)	91.96 (32.48%)	0.72 (0.25%)	173.78 (61.38%)	283.10 (100%)

NR-CVSS circuits. When applying the BLUD, we can increase the $f_{\max}$ by at least to 125 MHz. Fig. 13 presents the post-layout simulation results. Table III also presents the active power consumption of a read operation at 125 MHz when applying the BLUD. The power consumption of the BLLDs increases slightly due to a brief dc. Hence, the overall power consumption of the SRAM only increases marginally.

Table IV presents the post-layout simulation results of two 22-nm SRAMs designed with the same configuration and access bits but with different assist circuits. The reference SRAM is from the industry and adopts the WLUD for read assist and the NBL for write assist with the VDD of the memory core and peripheral circuits specified at $0.8 \text{ V} \pm 10\%$ and $0.7\text{--}0.8 \text{ V} \pm 10\%$, respectively. Although we cannot operate the industrial SRAM in a lower supply voltage for real applications, we performed post-layout simulations for the whole SRAM in 0.5 and 0.6 V to check the possible maximal performance, the access energy, and the power-performance-area (PPA) product. The PPA is calculated as “Energy/access-bit $\cdot$ area/(process) 2 /capacity.” The quantitative comparisons reveal the superiority of the proposed design techniques in LV operations. For example, the proposed SRAM improves the maximal frequency by 34% and reduces the access energy by 49% when VDD = 0.5 V. Although the

TABLE IV
POST-LAYOUT SIMULATION RESULTS OF SRAMs
UTILIZING DIFFERENT ASSIST SCHEMES

		Reference SRAM	This work
Capacity		256 Kb (same configuration and access bits)	
Bit cell		standard 6T	CA-6T
Assist circuits	Write	NBL	NR-CVSS
	Read	WLUD	NL-CVDD BLUD
$f_{\max}$	0.5 V	100 MHz (1)	134 MHz (1.34)
Energy/access-bit	0.5 V	233.3 fJ (1)	118.6 fJ (0.51)
$f_{\max}$	0.6 V	240 MHz (1)	306 MHz (1.28)
Energy/access-bit	0.6 V	283.9 fJ (1)	172.3 fJ (0.61)
Macro Area (mm^2)		0.053 (1)	0.065 (1.23)
PPA @ 0.5 V † (fJ-mm 2 /nm 2 /bit)		97 (1)	61 (0.63)

† Energy/access-bit $\times$ normalized-area ($\equiv$ macro-area/(process) 2 /capacity)

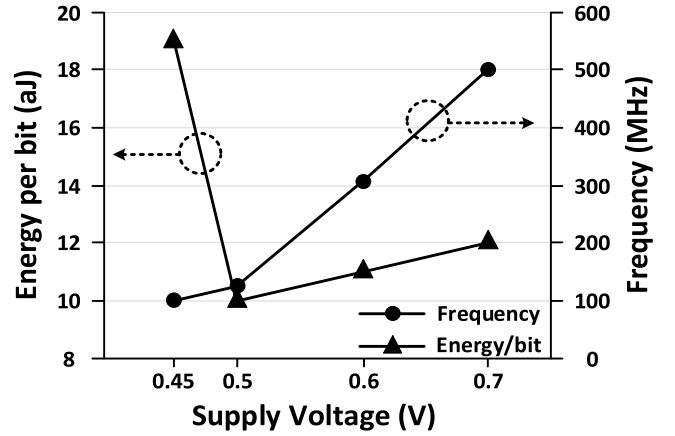


Fig. 15. Measurement results across 0.45–0.7 V.

proposed 0.5-V SRAM pays a 23% area penalty, it improves the PPA by 37%.

Fig. 15 depicts measurement results of the $f_{\max}$ and the active energy/bit of the SRAM, operating from 0.45 to 0.7 V for the typical corner chip with checkerboard patterns. The SRAM achieves a $f_{\max}$ of 125 MHz with 0.5 V/TT/25 $^\circ\text{C}$. The SRAM can operate down to 0.45 V with a $f_{\max}$ of 100 MHz. The index energy/bit reveals that the SRAM has a minimum energy point ($E_{\min}$) at 0.5 V.

Table V presents a comparison table of measurement results. We report energy consumption in two indices, energy/access-bit, as [20] did, and energy/total-bit, to reflect the leakage effect of un-accessed banks in a high-capacity SRAM. Though [17] and [20] look to achieve good energy indices, they are not face-to-face compared to this work because the power of bias circuits is not considered in the report. The normalization is made to [9], considering the completeness of the reported data.

The achieved $f_{\max}$ represents a 6.25 $\times$ improvement compared to the SRAM [9]. Measurement results verify the proposed high-speed LV design technique, that is, the CA-6 T bit cell utilizing the cooperated column-wise NL-CVDD assist circuit, row-wise NR-CVSS assist circuits, and BLUD BL assist circuit. Also, owing to the reduced voltage headroom by adopting the no-dc normally lowered CVDD and normally raised CVSS assist circuits, the achieved active energy at 0.5 V is only 10 aJ/bit, a 91% energy reduction compared to [9]. The

TABLE V
COMPARISON TABLE

		JSSC'17 [10]		JSSC'18 [18]	TCAS1'20 [7]	TVLSI'21 [8]	TCAS1'18 [17]	TCAS1'23 [20]	JSSC'22 [19]	TCAS1'17 [9]	This work			
Technology		28-nm Bulk CMOS		28-nm Bulk CMOS	28-nm Bulk CMOS	28-nm FDSOI	28-nm FDSOI	28-nm FDSOI	55-nm Bulk CMOS	28-nm Bulk CMOS	22-nm Bulk CMOS			
Capacity		256 Kb		32 Kb	32 Kb	16 Kb	128 Kb	128 Kb	8 Kb	256 Kb	256 Kb			
Access bits		64		32	8	8	64	32	16	64	32			
Bit cell	Type	DSC-6T		customized 6T	untypical 6T	9T	7T	7T-NDR	13T	standard 6T	CA-6T			
	Cell area (normalized to standard-6T)	1×		n.a.	1.5× (including overhead for VDD control)	1.65×	2.16x	2.95x	n.a.	2.26× (including overhead for local-sensing)	1×			
Assist	Write	CSVG	NBL	WL boosting	2 VDD supplies with read/write control	Data-aware	WBL & WWL boosting + body biasing [‡]	FBF [‡] + negative WWL [‡]	3T write port	Vtrip-tracking	NR-CVSS	NL-CVDD		
	Read		-	n.a.		3T read port	RWL boosting + body biasing [‡]	2T report	4T read port	4-bit Local BL		BLUD		
	Standby	n.a.		n.a.	n.a.	n.a.	Tail buffer + body biasing [‡]	Dual supply voltage + NDR + FBF [‡]	HVT transistors in 6T	Threshold power-gating	Peripheral: Super cut-off Cell array: NL-CVDD + NR-CVSS			
	Deep sleep	n.a.		n.a.	n.a.	n.a.	n.a.	Peripheral: Dual supply voltage + Power gating Cell array: NDR + ZBB [‡]	n.a.	n.a.				
	Shutdown	n.a.		n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.				
f_{max}	0.5 V or @ E_{min}	n.a.		n.a.	15 MHz ⁺ @ E_{min} 0.41 V	7.5 MHz ⁺	15 MHz @ E_{min} 0.24 V	64 MHz	Read: 4.3 MHz Write: 4.6 MHz	20 MHz (1)	125 MHz (6.25)			
Energy /access-bit					167.9 fJ @ 0.41 V	840 fJ	36 fJ ^{&} @ 0.24 V	24.5 fJ ^{&}	Read: 18 pJ Write: 31 pJ	468 fJ (1)	81.9 fJ (0.18)			
Energy /total-bit					41 aJ @ 0.41 V	410 aJ	17.58 aJ ^{&} @ 0.24 V	6 aJ ^{&}	Read: 35.1 fJ Write: 60.6 fJ	114 aJ (1)	10 aJ (0.09)			
f_{max}	0.6 V or @ V_{min}	400 MHz*, <200 MHz [‡]		50 MHz @ V_{min} 0.7 V	n.a.	15 MHz ⁺	n.a.	120 MHz ⁺	Read: 21.9 MHz Write: 24.5 MHz	150 MHz (1)	305 MHz (2.03)			
Energy /access-bit		n.a.		n.a.				1.35 pJ ⁺	29.5 fJ ⁺ &	Read: 24.5 pJ Write: 43.4 pJ	94.2 fJ (1)	90.1 fJ (0.96)		
Energy /total-bit		n.a.		n.a.				659 aJ ⁺	7 aJ ⁺ &	Read: 47.9 fJ Write: 84.7 fJ	23 aJ (1)	11 aJ (0.48)		
SB Power /bit	0.5 V	n.a. (not available)		n.a.	n.a.	n.a.	2.5 pW ⁺ &	36 pW ^{&}	6.33 pW @ 0.4 V	229 pW (1)	39 pW (0.17)			
	0.6 V						5.25 pW ⁺ &	n.a.	n.a.	343 pW (1)	62 pW (0.18)			
DS Power /bit	0.5 V						n.a.	0.04 pW ^{&}		n.a.	n.a.			
	0.6 V													
SD Power /bit	0.5 V						n.a.						n.a.	
	0.6 V	n.a.		17 pW										
Macro Area (mm ²) (Normalized [‡]) (percentages of assists)		n.a.		n.a.	0.028 (0.56)	n.a.	0.161 ^{&} (0.81)	0.075 ^{&} (0.38)	0.041 (0.86)	0.397 (1.00)	0.065 (0.27) (CVSS/CVDD/BLUD: 7.2%/6.7%/2.9%)			
PPA [‡] (fJ-mm ² /nm ² /bit)		n.a.		n.a.	183 @ 0.41 V	n.a.	56 ^{&} @ 0.24 V	18 ^{&} @ 0.5 V	Read: 29781 Write: 51290 @ 0.5 V	904 @ 0.5 V	42 @ 0.5 V			

* Measured with off-chip CVSS [‡] Simulation results of redesigned DSC-6T SRAM macro with CSVG [‡] From outside of the SRAM [‡] Estimation

[‡] SRAM macro-area/(process)²/capacity [‡] Energy/access-bit × normalized-area [‡] Not include power and area of some assist voltage generators

CVDD and CVSS assist circuits contribute to a significant 83% power reduction at 0.5 V in SB mode compared to [9].

Most conventional LV SRAMs lack the design of power-saving modes. In DS mode, the proposed SRAM dissipates

only 12 pW/bit at 0.5 V by applying the super-cut-off switches to the peripheral circuits. When the CVDD circuit turns to a super cut-OFF state to shut down cell arrays, the SRAM dissipates only 10 pW/bit at 0.5 V in SD mode. Finally,

the proposed SRAM has a 73% area reduction compared to the state-of-the-art design [9]. We can lower the VDD for the cell arrays to reduce the leakage power even more. The measured minimum retention voltage is 0.4 V, and the total deep-sleep power at the deep-sleep mode is 2 μ W, amounting to 7.67 pW/bit power efficiency (not included in the table).

VI. CONCLUSION

This work proposes coordinated assist circuits for the LV SRAM design. Dynamic noise-margin analyses derive the design guidelines of assist voltages. The implemented 22-nm 256-Kb SRAM can operate down to 0.45 V, with a voltage of the minimum energy point at 0.5 V. At 0.5 V/TT/25 °C, the measured maximum frequency is 125 MHz with a 10 aJ/bit active energy, representing a 6.25 $\times$ frequency improvement with a 91% energy reduction, compared to the state-of-the-art 0.5 V SRAM. The SRAM also achieves a 12 pW/bit deep-sleep power and a 10 pW/bit SD power at 0.5 V.

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